

From Support to State

Auditing Structural and Dynamic Descent in the Male Drosophila CNS

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This paper is Paper XVI of the RIME program, an independently scoped computational connectome case study. It does not extend or reopen the SOF protocol line closed by Paper XV.

Abstract

Problem. Large connectomes are commonly compressed into sector-level support graphs, signed operators, coarse state variables, and low-dimensional readouts. These representations answer different questions. A path in a quotient support graph need not lift to a nonzero microscopic composition, and a reproducible coarse observable need not determine its own future.

Approach. The analysis separates three descent problems on the MaleCNS v1.0 connectome: structural support-to-route descent, operator-to-observation visibility, and microscopic-to-coarse dynamic closure. Exact finite route audits are combined with source-bound numerical dynamics on a frozen signed, normalized carrier. Deterministic fiber criteria distinguish descriptive observability, predictive fit, and exact closure.

Results. At depth two, the typed-core, relay-strict, and coverage-inclusive sectorizations lift 4,970 of 6,237, 5,751 of 7,019, and 6,125 of 7,831 quotient-supported triples, respectively. The first nested increment lifts 781 of 782 newly admitted triples, whereas the second lifts 374 of 812. Matched-support transmitter-proxy sign assignments produce much larger registered response changes than the corresponding coverage loss, and the visibility of a fixed perturbation varies strongly with the observation functional. For dynamic descent, both superclass-mean and somaSide-mean observations fail the registered first-order fiber criterion; one coarse lag also fails, including an exact witness in the named `cb_intrinsic` sector. On the same signed carrier used by the dynamic test, somaSide has exact complete liftability at depths two and three, yet its first-order coarse dynamics do not close. This is a bounded same-carrier computational counterexample to the promotion from finite static liftability to deterministic dynamic closure.

Boundary. The paper reports exact finite certificates and bounded computational observations. It does not establish physiological excitation/inhibition, causal signal propagation, biological competence, partition superiority, all-depth liftability, or closure at higher memory orders.

Keywords. connectome; coarse-graining; quotient support; routed composition; lumpability; dynamic closure; representation loss; MaleCNS

Notation Table

symbol	meaning
V	finite node set of the registered carrier
$V = \bigsqcup_{i \in I} V_i$	declared sectorization
Q_i	coordinate projector onto sector V_i
Y	registered microscopic operator or carrier matrix
$\text{Path}_d^{\text{quot}}$	depth- d paths admitted by consecutive quotient support
$\text{Route}_d[Y]$	depth- d sector words with nonzero projected product
F_Y	registered deterministic microscopic update under carrier Y
O	declared coarse observation map
$\bar{F}$	putative deterministic coarse update
$\Psi_1(x)$	one-lag coarse history $(O(F_Y x), O(x))$
LP_d	finite depth- d liftability proportion
$Y_{\text{known}+}$	known-sign support with positive weights
$Y_{\text{known}\pm}$	signed, normalized transmitter-proxy carrier
<code>superclass, somaSide</code>	registered MaleCNS sectorizations

1 Introduction

A connectome is simultaneously a graph, a weighted operator, a source of candidate routes, and a substrate for dynamical models. These roles are often compressed into one informal statement: if two coarse regions are connected, information can flow between them, and if their aggregate activity can be observed, that activity can be evolved as a coarse state. Neither promotion is automatic.

The first issue is compositional. A quotient support graph forgets which microscopic relay realizes each adjacent sector transition. Consecutive quotient edges can therefore be supported by incompatible relays. Even when a microscopic route exists, signed route families may cancel in the projected operator product. The correct object is not merely a graph path but a carrier-qualified routed product.

The second issue is dynamical. An observation $O(x)$ may be well-defined, reproducible, and useful for description while failing to determine $O(F_Y x)$. In that case the observation is not a deterministic state for the registered dynamics. Prediction quality and exact factorization are also different: a fitted predictor may approximate an output without making the microscopic update constant on observation fibers.

This paper audits these distinctions on the MaleCNS v1.0 connectome [1]. The dataset provides a whole adult male central nervous system spanning brain and nerve cord. Its scale makes representation loss unavoidable, but scale alone does not determine which representations preserve which structures. This leads to one controlled question:

Which structural information survives successive coarse representations strongly enough to remain compositional, dynamically consequential, observable, and state-sufficient?

The audit follows the chain

$$\boxed{\begin{array}{l} \text{support} \longrightarrow \text{routed composition} \longrightarrow \text{signed operator} \\ \longrightarrow \text{dynamics} \longrightarrow \text{observation} \longrightarrow \text{dynamic closure} \end{array}} \quad (1.1)$$

Each arrow requires its own certificate. Failure at one arrow does not erase validity at an earlier layer, and success at an earlier layer does not certify a later one. Figure 1 summarizes the typed audit.

Each arrow requires an explicit descent certificate; no arrow is automatic.

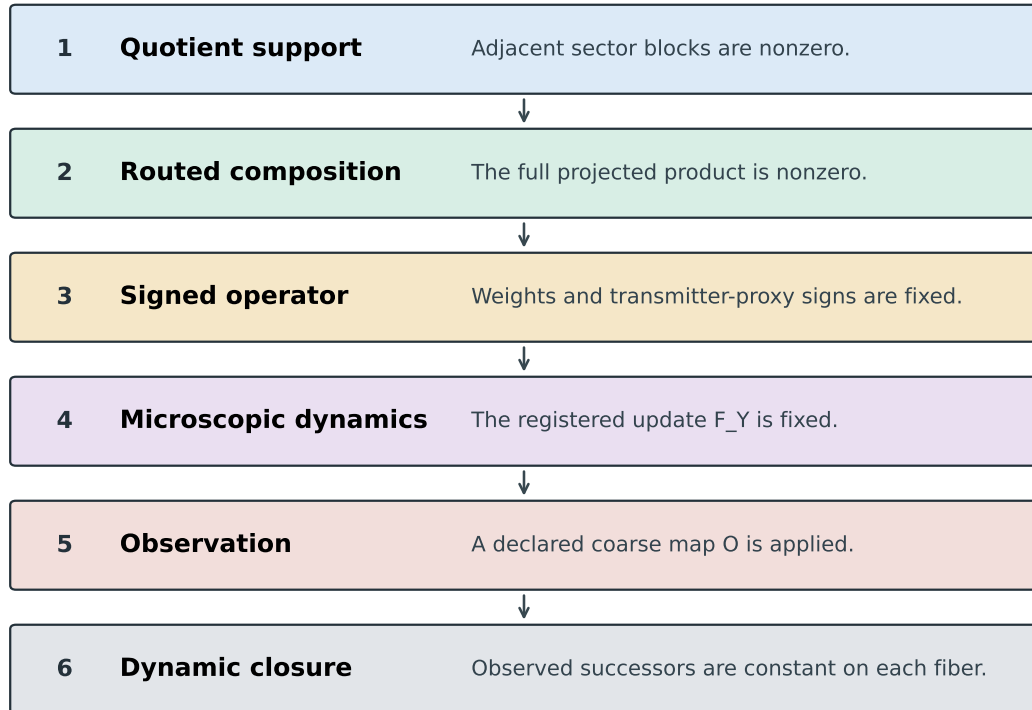


Figure 1. The six-layer descent stack audited in this paper. Each arrow requires an explicit descent certificate; no arrow is automatic.

1.1 Contributions

The paper makes four contributions.

1. **A common descent language.** Three elementary propositions formalize route-to-path inclusion, deterministic fiber closure, and one-lag history closure. They provide exact gates for the empirical audits.
2. **A structural support audit.** Three MaleCNS sectorization policies show that coverage admission and relay admission are distinct. The strongest nested contrast is 781/782 versus 374/812 lifted additions.
3. **A signed dynamic and observation audit.** Matched-support sign changes and alternative observation functionals are shown to affect different layers of the registered response.
4. **A same-carrier counterexample.** On $Y_{\text{known}\pm}$, somaSide has complete exact liftability at depths two and three while failing deterministic first-order dynamic closure. Static liftability and dynamic state sufficiency therefore require separate certificates even on the same carrier and partition.

1.2 Nonclaims

The analysis does not infer physiological excitation or inhibition from the transmitter proxy, does not identify causal biological circuits, and does not model living-fly behavior. It does not rank sectorizations by the magnitude of their raw closure defects. It does not claim all-depth liftability, closure or nonclosure at memory order $k \geq 2$, or a universal theorem about biological coarse-graining.

2 Related Work and Novelty Boundary

State aggregation and lumpability ask when a projected process retains a closed stochastic description. Classical finite-state treatments formulate first-order lumpability through consistency of transition probabilities on partition blocks [8]. Higher-order lumpability separates first-order Markov closure from closure after retaining a finite observation history [7]. The deterministic fiber criteria used below are the corresponding elementary factorization statements for a fixed map and a declared observation domain. This paper does not claim lumpability theory as new.

Exact coarse-graining can also be recovered by choosing collective variables designed to preserve particular dynamical quantities. Lu and Vanden-Eijnden, for example, construct collective variables that preserve transition-order and first-passage information without assuming time-scale separation [10]. The question here is different: the observation maps are fixed biological annotations, and the audit asks whether those already declared maps satisfy exact finite factorization criteria.

Network quotient constructions likewise require compatibility between graph structure and dynamics. Graph fibrations can induce exact maps between networked dynamical systems under declared structural hypotheses [5], while connectome coarse-graining studies typically test which dynamical features survive a selected aggregation [9]. These results motivate, but do not replace, the carrier-specific route and fiber checks performed here.

The MaleCNS release supplies the anatomical substrate and annotation surface [1, 11]. Neurotransmitter labels derived from electron microscopy are valuable connectomic annotations, but their status is predictive and does not by itself determine receptor-specific postsynaptic

effect [6]. Accordingly, $Y_{\text{known}\pm}$ is called a transmitter-proxy signed carrier, not a physiological excitation/inhibition matrix.

Within the RIME program, Paper VIII distinguishes labelled operator carriers from quotient support and closure objects [4], Paper XX develops all-depth carrierwise route criteria [2], and Paper XXIV states the general discipline that structure descends through a lossy map only under an explicit compatibility condition [3]. The present paper contributes a large-connectome case study with two features not supplied by those abstract interfaces: source-bound MaleCNS computations and a same-carrier separation between complete finite static liftability and failed deterministic dynamic closure.

The novelty claim is therefore deliberately narrow. It is not a new general theory of lumpability, graph quotienting, or connectome dynamics. The contribution is a typed, source-addressed audit that keeps structural support, routed composition, signed dynamics, observation visibility, and state closure from being silently identified.

3 Typed Descent Criteria

3.1 Quotient Paths and Routed Products

Let $V = \bigsqcup_{i \in I} V_i$ be a finite partition with coordinate projectors Q_i , and let $Y \in \text{End}(\mathbb{K}^V)$. A sector word $\mathbf{i} = (i_0, \dots, i_d)$ is quotient-supported when every consecutive block is nonzero:

$$Q_{i_r} Y Q_{i_{r-1}} \neq 0, \quad 1 \leq r \leq d.$$

It is routed-active when the full projected product is nonzero:

$$Q_{i_d} Y Q_{i_{d-1}} Y \dots Y Q_{i_0} \neq 0.$$

Write the corresponding sets as $\text{Path}_d^{\text{quot}}$ and $\text{Route}_d[Y]$.

Proposition 3.1 (Route-to-path inclusion).

For every finite declared sectorization and operator Y ,

$$\text{Route}_d[Y] \subseteq \text{Path}_d^{\text{quot}}.$$

Equality requires an additional carrier-qualified lifting condition.

Proof. If one consecutive factor $Q_{i_r} Y Q_{i_{r-1}}$ vanishes, then the full product containing that factor vanishes. Hence every active routed word is quotient-supported. The reverse implication can fail because the images and kernels of adjacent blocks need not align. Over signed carriers, distinct microscopic contributions can also cancel.

For nonnegative adjacency data, nonzero matrix entries can be interpreted as microscopic chains, so failure typically comes from incompatible relays. For signed operators, Boolean route existence is only an intermediate check: exact projected accumulation is still required. $\square$

3.2 Deterministic Dynamic Closure

Let X be a microscopic state set, $F : X \rightarrow X$ a deterministic update, and $O : X \rightarrow Z$ an observation map. The map O closes at first order on a declared domain $D \subseteq X$ when there is a map $\bar{F} : O(D) \rightarrow Z$ such that

$$O(Fx) = \bar{F}(Ox), \quad x \in D. \tag{3.1}$$

Proposition 3.2 (Deterministic fiber criterion).

A map $\bar{F}$ satisfying (3.1) exists if and only if

$$O(x) = O(y) \implies O(Fx) = O(Fy) \quad (x, y \in D). \quad (3.2)$$

Proof. If $\bar{F}$ exists, (3.2) follows by applying it to the common observation. Conversely, define $\bar{F}(z) = O(Fx)$ for any $x \in D$ with $O(x) = z$. Condition (3.2) makes this definition independent of the chosen representative.

A pair violating (3.2) is an exact obstruction to deterministic first-order closure on the declared domain. It is not merely a prediction error. A small numerical discrepancy may still require tolerance analysis, but once the source states are exactly co-observed and their successors differ under the registered arithmetic, no deterministic factorization exists on that domain. $\square$

3.3 One-Lag Closure

Define the one-lag history map and its target by

$$\Psi_1(x) = (O(Fx), O(x)), \quad T_2(x) = O(F^2x).$$

Proposition 3.3 (One-lag history-fiber criterion).

There exists a map $G : \Psi_1(D) \rightarrow Z$ satisfying $T_2 = G \circ \Psi_1$ on D if and only if

$$\Psi_1(x) = \Psi_1(y) \implies T_2(x) = T_2(y) \quad (x, y \in D). \quad (3.3)$$

Proof. This is Proposition 3.2 applied to the source map Ψ_1 and target T_2 .

Failure of (3.3) proves only that one retained lag is insufficient on the declared domain. It says nothing about closure at order $k \geq 2$, stochastic closure, or approximate predictive adequacy.

The three propositions separate the paper’s object types:

quotient visibility $\neq$ routed composition $\neq$ dynamic state sufficiency.

(3.4)

$\square$

4 MaleCNS Data and Registered Objects

4.1 Dataset and Provenance

The source dataset is MaleCNS v1.0, released as a complete adult male *Drosophila* central nervous system connectome with brain and nerve cord annotations [1, 11]. Its registered neuPrint identifier is `male-cns:v1.0`. The paper-owned provenance record binds the annotation table, neurotransmitter table, and directed table of body-pair connection counts by exact file size and SHA-256. The dataset license is recorded separately from the repository code license.

The computations use body-pair detected-connection counts rather than a claim of biological synaptic efficacy. Annotation values, including `<missing>`, are data values in a declared operational partition. They are not silently imputed. The named-sector follow-up in Section 7 explicitly separates a biologically named superclass from the missing-label bucket.

4.2 Static and Dynamic Carriers

The initial static audit uses a nonnegative MaleCNS adjacency carrier to test whether quotient-supported sector words have microscopic relay realizations. The dynamic line constructs a node-level carrier and then a normalized signed operator $Y_{\text{known}\pm}$ from the registered known-sign subset. The signed labels are transmitter proxies. They do not encode receptor composition, co-transmission, or postsynaptic physiological sign.

The stored operator uses source rows and target columns. State propagation is therefore the incoming-action convention $x \mapsto Y_{\text{known}\pm}^\top x$; routed products retain the corresponding source-to-target order. This orientation is part of the registered carrier and is not interchangeable with left multiplication by the stored matrix.

Registered dynamics. Let W denote the frozen signed known-transmitter matrix and set

$$s_u = \sum_v |W_{uv}|, \quad Y_{uv} = \begin{cases} W_{uv}/s_u, & s_u > 0, \\ 0, & s_u = 0. \end{cases}$$

The deterministic update used in the dynamic audits is

$$F_Y(x) = (1 - \alpha)x + \alpha \phi(\gamma Y^\top x), \quad \phi(z)_v = \min\{x_{\max}, \max\{0, z_v\}\}. \quad (4.1)$$

The registered values are $\alpha = 0.2$, $\gamma = 1$, and $x_{\max} = 1$, with zero external input and no stochastic noise. For a declared partition $V = \bigsqcup_i V_i$, the observation is the sector-mean vector

$$O_i(x) = \frac{1}{|V_i|} \sum_{v \in V_i} x_v. \quad (4.2)$$

The dynamic carrier admits 211,577 nodes and 26,028,386 directed edges in its full support. The known-sign support used by $Y_{\text{known}\pm}$ contains 25,214,058 directed edges, or 96.871385% of admitted edges, and covers 97.929437% of raw-weight mass. The matched-support positive control $Y_{\text{known}+}$ retains the same support and absolute edge magnitudes while removing the registered sign assignment.

4.3 Sectorizations and Observations

Two partitions are central:

- **superclass**, a 28-dimensional operational annotation partition in the dynamic comparison;
- **somaSide**, a four-dimensional side-based partition.

For a partition $V = \bigsqcup_i V_i$, the dynamic observation is the vector of sector means. These means are descriptive estimands. Their existence and reproducibility do not imply Proposition 3.2.

4.4 Evidence Classes

The exact route audits and the named-sector finite history witness are Computational Certificates: finite enumerations bound to declared inputs and exact arithmetic checks. The signed dynamics, observation-resolution ratios, and registered closure defects are Computational Observations. The release validator performs local closure verification; it does not constitute an independent validation of the scientific model.

5 Static Structural Descent

5.1 Coverage Admission Is Not Relay Admission

The depth-two audit compares three nested sector-universe policies. A quotient-supported triple is counted as lifted only when the declared microscopic relay condition is satisfied.

policy	quotient-supported triples	lifted triples	liftability
typed-core	6,237	4,970	0.7969
relay-strict	7,019	5,751	0.8193
coverage-inclusive	7,831	6,125	0.7821

The nested additions are more informative than comparing the three ratios as if they were estimates on a common denominator:

enlargement	newly supported	newly lifted	conditional liftability
typed-core to relay-strict	782	781	99.87%
relay-strict to coverage-inclusive	812	374	46.06%

The first enlargement almost always contributes usable relays. The second adds many labels that increase coverage without supplying compatible microscopic composition. Thus

$$\boxed{\text{coverage-admissible} \neq \text{relay-admissible.}} \quad (5.1)$$

This is a finite audit of the registered carrier and policies. It is not a universal statement about annotation systems, and the observed percentages are not biological transmission probabilities.

5.2 Exact Signed Liftability on the Dynamic Carrier

The static/dynamic comparison must use the same carrier. The follow-up audit therefore re-computes route activity directly on $Y_{\text{known}\pm}$ rather than importing nonnegative relay existence. Its decision hierarchy is:

1. a unique microscopic route gives exact nonzero activity;
2. a sign-homogeneous route family gives exact nonzero activity;
3. otherwise exact rational accumulation and modular checks decide the projected product;
4. absence of a microscopic route is recorded separately from cancellation.

For somaSide, every quotient-supported word is active:

depth	quotient-supported words	exact nonzero routes	LP_d
2	64	64	1
3	256	256	1

For superclass, the corresponding counts are 3,667/4,783 at depth two and 41,247/67,562 at depth three. Every zero in the superclass audit is classified as absence of a microscopic route; no unresolved or cancellation zero remains in the registered result.

The somaSide statement is complete only at the two audited depths. It does not imply $LP_d = 1$ for arbitrary d .

6 Signed Semantics and Observation Resolution

6.1 Matched Support, Different Signed Dynamics

The signed-dynamics control separates support loss from sign assignment:

$$Y_{\text{full}+} \longrightarrow Y_{\text{known}+} \longrightarrow Y_{\text{known}\pm}.$$

The second transition preserves support and absolute magnitudes, changing only the registered transmitter-proxy signs. Across the frozen gain grid, the cosine distances between $Y_{\text{known}+}$ and $Y_{\text{known}\pm}$ responses are 0.84668, 0.87846, and 0.90458 for $\gamma = 0.9, 1.0, 1.1$. The corresponding coverage-reference distances between $Y_{\text{full}+}$ and $Y_{\text{known}+}$ are 0.00416, 0.00598, and 0.00770.

This establishes a model-relative separation: on the registered grid, the matched-support sign assignment changes response allocation much more than the registered coverage loss. It does not validate the proxy as physiological excitation/inhibition, nor does it identify a causal circuit.

6.2 The Observation Functional Changes Visibility

The observation-resolution audit retains the registered node-level paired trajectory and changes only the readout. Mean, RMS, 95th-quantile, and local observations expose substantially different magnitudes of the same model-relative perturbation.

source stratum	RMS / mean	q95 / mean	local / mean
small	8.7141	2.0207	24.0335
medium	7.0140	1.5190	3.8780
large	8.7840	2.2083	36.4267

The comparison does not privilege one observation as biologically correct. It shows that effect visibility is a property of the pair

(registered dynamics, observation functional),

not of the microscopic perturbation alone. Figure 2 collects the principal registered contrasts.

Frozen MaleCNS representation-loss contrasts

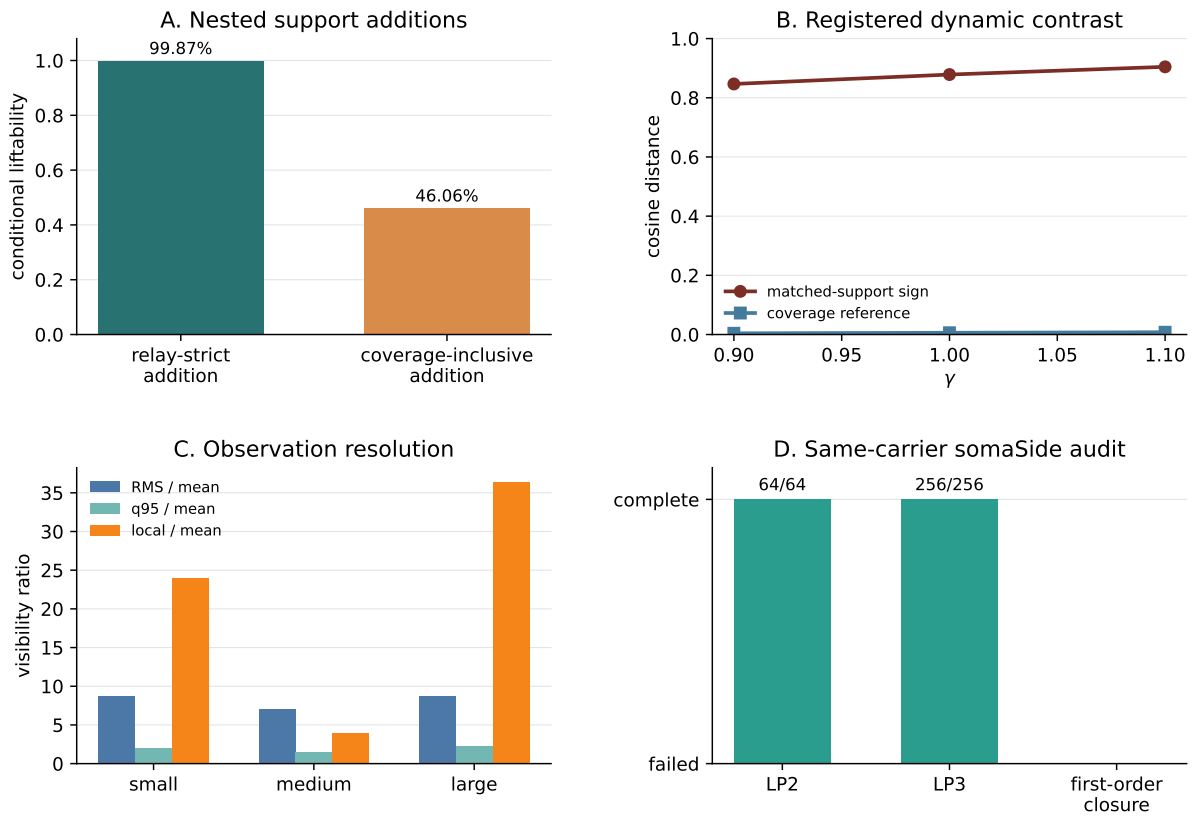


Figure 2. Principal frozen contrasts. Values are descriptive coordinates from the declared audits; bars across different panels do not share a scientific unit or denominator.

7 Dynamic Closure and Memory

7.1 First-Order Fiber Failure

The D2.1 protocol fixes $Y_{\text{known}\pm}$, the superclass-mean observation, $\gamma = 1$, $\alpha = 0.2$, zero input, and a unit-state first-reference domain. It then tests Proposition 3.2 directly. The result is

FAILED_FIRST_ORDER_MARKOV_CLOSURE.

More explicitly, with e_a the unit state at node a and s_a the absolute outgoing-row mass defined in Section 4.2, the declared domain is

$$D = \{e_a : s_a > 0\}. \quad (7.1)$$

The fiber key is the superclass containing a , since all unit states in the same sector have the same $O(e_a)$. Within each sector, candidate nodes are scanned in increasing registered node index; the first candidate is retained as the reference and every later candidate is compared only with that reference. Consequently the witness count below is a count of detected first-reference inconsistencies, not a count of all hostile pairs.

The finite scan records 141,296 first-reference inconsistency witnesses and a primary reference L1 defect of 0.005651574473947287. The count is not an obstruction prevalence estimate, and the defect is a witnessed lower bound, not a complete fiber diameter.

The conclusion is exact at the logical level of the registered witness: superclass mean is not a deterministic first-order state on the declared domain. The numerical magnitude is not a measure of biological importance.

7.2 One Coarse Lag Still Fails

The D2.2 protocol tests Proposition 3.3 with

$$\Psi_1(x_0) = (z_1, z_0), \quad T_2(x_0) = z_2.$$

The history equality gate is exact. For $e_a \in D$, define the positive mass sent from a into target sector V_j by

$$p_{aj} = \sum_{v \in V_j: W_{av} > 0} W_{av}, \quad g_a = \gcd(s_a, p_{a1}, \dots, p_{am}).$$

If $i(a)$ is the source-sector index, the canonical history signature is the integer tuple

$$h(a) = \left(i(a), \frac{s_a}{g_a}, \frac{p_{a1}}{g_a}, \dots, \frac{p_{am}}{g_a} \right). \quad (7.2)$$

Thus equality is tested by reduced integers, without floating-point tolerance; the computed trajectories are used only as a numerical cross-check and to test successor separation. This signature produces 119,138 groups, of which 5,745 contain collisions, comprising 50,046 states. The frozen first witness has a history cross-check error of $2.117582368135751 \times 10^{-22}$ and a successor L1 separation of $2.567412844075076 \times 10^{-7}$.

That first witness lies in the operational <missing> superclass bucket. To separate closure failure from a missing-annotation artifact, an additive hostile follow-up preserves the original receipt and searches only named sectors in deterministic order. It finds an exact history-fiber witness in `cb_intrinsic` after examining 5,367 ordered candidate pairs. The two source states have identical registered one-lag histories and differ in nine exact coordinates of the next observation.

The follow-up strengthens the partition-level obstruction. It remains an operational annotation result, not a biological mechanism claim. No test of $k \geq 2$ is reported.

7.3 Representation-Relative Failure

The D2.3 protocol repeats the first-order test on the same carrier and dynamic settings for superclass mean and somaSide mean.

observation	dimension	collision		reference L1 defect
		fibers	witnesses	
superclass mean	28	22 / 25	141,296	0.005651574473947287
somaSide mean	4	4 / 4	163,435	0.0004874856094383272

The collision fibers contain 163,436 and 163,439 states, respectively. Both observations fail first-order closure. The raw defect magnitudes cannot rank the partitions: their dimensions, sector-size profiles, compression ratios, and fiber geometries differ. A no-witness result under either scan would have meant only that no obstruction was found on the registered domain, not a proof of closure.

8 Same-Carrier Static and Dynamic Separation

The strongest cross-layer result combines Sections 5.2 and 7.3 without changing the carrier or partition:

$$\begin{aligned}
 Y &= Y_{\text{known}\pm}, \\
 \text{LP}_2(\text{somaSide}; Y) &= 64/64 = 1, \\
 \text{LP}_3(\text{somaSide}; Y) &= 256/256 = 1, \\
 \text{somaSide first-order closure under } F_Y &= \text{FAILED}.
 \end{aligned} \tag{8.1}$$

Here “same carrier” means the same frozen node basis, edge weights, signs, and outgoing-row normalization. Routed products use the registered source-row/target-column storage convention for Y , whereas F_Y uses the registered transpose action in (4.1). These are two declared uses of the same operator bytes, not different carriers.

Computational Certificate 8.1 (Signed liftability). On the frozen signed, normalized $Y_{\text{known}\pm}$ carrier, every somaSide quotient-supported word of depths two and three has an exact nonzero projected product.

Computational Observation 8.2 (Dynamic fiber obstruction). On the same carrier and somaSide partition, the registered first-order dynamic map is not constant on observation fibers.

Together they yield the bounded same-carrier statement:

$$\boxed{\text{perfect static liftability at depths 2 and 3} \not\Rightarrow \text{deterministic first-order dynamic closure.}} \tag{8.2}$$

The antecedent is finite-depth and the consequent is domain-qualified. Thus (8.2) is a computational counterexample to a particular promotion, not a general theorem that static liftability and dynamic closure are unrelated. The two properties can coexist under stronger hypotheses; they simply do not follow from one another here.

The reason is structural. Static liftability asks whether declared finite sector words have nonzero microscopic realization. Dynamic closure asks whether all microscopic states identified

by an observation have identical observed successors. The latter depends on amplitudes within observation fibers, not only on the existence of short inter-sector routes.

9 Discussion

9.1 Observability, Predictability, and Closure

The case study supports a three-way distinction:

$$\boxed{\text{observability} \neq \text{predictability} \neq \text{closure.}} \quad (9.1)$$

Observability means that a declared statistic can be computed from the state. Predictability concerns the performance of a model under a declared loss and distribution. Closure is an exact factorization statement. A predictive model may be useful in the absence of closure, while a closure theorem requires the fiber condition regardless of empirical fit.

This distinction matters for connectome-scale modeling because annotations are often designed for biological description rather than dynamical sufficiency. The failure of superclass or somaSide mean to close does not make those annotations invalid. It states only that additional state, memory, stochasticity, or approximation is needed if they are used as autonomous dynamic variables.

9.2 Static Support Is an Observation, Not a Dynamics

A quotient graph is a valuable summary of possible adjacency. Its paths are not automatically compositional routes, and compositional routes are not automatically a closed coarse dynamics. The MaleCNS audits display both gaps: coverage-inclusive support admits many nonlifting triples, while complete somaSide lifting at two finite depths does not repair the dynamic fiber obstruction.

The general methodological rule is:

Never infer preservation of a structure through a lossy representation without a descent theorem or a carrier-qualified certificate.

This rule is not anti-compression. It identifies what must accompany a compression if a particular downstream operation is to remain meaningful.

9.3 Sign and Readout Are Separate Modeling Choices

The matched-support sign control shows that identical graph support and edge magnitudes do not determine the registered dynamic response. The observation audit then shows that one node-level response does not determine its measured visibility independently of the readout. These are separate modeling layers:

$$\text{support} \quad | \quad \text{operator semantics} \quad | \quad \text{observation semantics.}$$

The vertical bars are type boundaries, not an order relation. Each layer can be varied while holding part of the others fixed.

9.4 Limitations

The study uses one connectome release, registered annotation partitions, one family of deterministic dynamics, and finite test domains. Transmitter labels are proxies; receptor-specific sign and neuromodulatory effects are absent. The closure scans are not prevalence estimators. The finite route certificate stops at depth three, and the memory audit stops at one lag. The paper makes no claim that the tested dynamics are a faithful behavioral model of the fly.

The strongest result is therefore methodological and object-relative: static liftability, signed response, observation visibility, and deterministic closure are distinct properties, and the registered MaleCNS carrier supplies explicit controls separating them.

10 Claim Status and Boundary

ID	statement	evidence level	boundary
C1	Quotient support can overstate compositional realizability.	Computational Certificate	Registered finite carrier and depths only.
C2	Coverage admission and relay admission differ.	Computational Certificate	No causal or physiological interpretation.
C3	Matched-support transmitter-proxy signs alter the registered response.	Computational Observation	Not receptor-validated E/I.
C4	Observation functional changes visibility of a fixed model-relative effect.	Computational Observation	No universally privileged readout.
C5	Registered superclass mean fails deterministic first-order closure.	Computational Observation	Declared domain and tolerance only.
C6	One lag does not restore closure; a named-sector exact witness exists.	Computational Certificate	No claim for $k \geq 2$ or all finite memory.
C7	Both tested mean representations fail first-order closure.	Computational Observation	Raw defects do not rank partitions.
C8	On $Y_{\text{known}\pm}$, somaSide has exact $\text{LP}_2 = \text{LP}_3 = 1$ while first-order closure fails.	Computational Observation	Finite-depth, same-carrier counterexample only.

The overall paper remains a **Computational Observation** because its main cross-layer conclusion consumes both exact finite certificates and model-relative numerical dynamics.

No result establishes physiological excitation/inhibition, biological signal propagation, causal necessity, behavior, learning, intelligence, consciousness, partition optimality, or a universal coarse-graining theorem.

11 Conclusion

The MaleCNS case study separates five objects that are easy to conflate: quotient support, microscopic routed composition, signed operator semantics, observed response, and closed coarse state. Exact path lifting and deterministic fiber factorization provide different gates because they preserve different structures.

The finite audits show that coverage can exceed compositional relay support, matched support can conceal large signed dynamic differences, and a fixed node-level effect can appear very different under alternative readouts. Most decisively, somaSide is perfectly liftable through depths two and three on the same signed carrier for which its first-order observed dynamics fail to close.

The resulting principle is simple: a lossy representation may remain valid for one purpose and fail for another. The correct response is neither to reject coarse representations nor to treat them as autonomous by default, but to state the structure being preserved and require the corresponding descent certificate.

Appendix A Computational Artifacts

The paper-owned release closure is under [experiments/paper16/](#) in the [RIME repository](#). The exact source and producer subset consumed by this paper is published under `source/`; unrelated exploratory work is not part of the release identity. The three upstream MaleCNS tables remain external official inputs bound by URL, byte size, and SHA-256. Large derived carrier arrays are rebuilt in scratch and compared with their registered byte identities.

role	short path
selected source and producers	<code>source/</code>
portable scientific artifacts	<code>source/**/results/</code>
static-audit replay receipt	<code>results/static-audits-replay.v1.receipt.json</code>
dynamic-descent replay receipt	<code>results/dynamic-descent-replay.v1.receipt.json</code>
exact A1/A2 replay receipt	<code>results/a1-a2-exact-replay.v1.receipt.json</code>
upstream dataset provenance	<code>upstream-provenance.v1.json</code>
release manifest	<code>release-manifest.v1.json</code>

The release manifest binds exact paths, byte sizes, SHA-256 digests, roles, and claim ownership. Large carrier outputs are referenced through a clean-clone producer-replay receipt. Portable artifact regeneration replaces machine-local locator strings with paper-owned relative paths without changing the scientific payload.

The validation graph is acyclic:

source and producer $\longrightarrow$ artifact $\longrightarrow$ receipt $\longrightarrow$ release manifest $\longrightarrow$ external anchor.

The release-closure receipt performs local closure verification. Separate paper-owned receipts record clean scratch replay and exact-byte agreement for the six static artifacts, six v0.1–v0.2 dynamic-descent artifacts, and the corrected A1/A2 producers. None of these receipts independently validates the scientific semantics, and no receipt occurs in its own transitive closure. Embodied finite-null results and inverse-identifiability experiments are outside this paper’s release identity.

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